

Essential Linux Commands

A Practical Guide with Real Outputs

This document provides a foundational reference for navigating and managing a Linux environment. It covers file operations, system monitoring, and network diagnostics.

1. pwd (Print Working Directory)

Outputs the absolute path of the current directory. This is useful for confirming exactly where you are located within the file system tree.

```
user@ubuntu:~$ pwd
/home/user
```

2. ls -la (List Directory Contents)

Lists all files and directories in the current location, including hidden files (those starting with a dot), along with their permissions, ownership, size, and modification date.

```
user@ubuntu:~/projects$ ls -la
total 24
drwxr-xr-x  3 user user 4096 May 16 09:00 .
drwxr-xr-x 15 user user 4096 May 15 18:30 ..
-rw-r--r--  1 user user  220 May 16 08:45 .bash_logout
drwxr-xr-x  2 user user 4096 May 16 09:05 src
-rw-r--r--  1 user user 1536 May 16 09:10 main.py
-rw-r--r--  1 user user  120 May 16 08:50 README.md
```

3. cat (Concatenate and Print)

Reads the contents of one or more files and prints them to the terminal screen. Often used for quickly reading configuration files or logs.

```
user@ubuntu:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.2 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.2 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
```

4. grep (Global Regular Expression Print)

Searches for a specific pattern of text within a file or stream. In this example, we search for the string "Error" inside a system log file.

```
user@ubuntu:/var/log$ grep "Error" syslog
May 16 10:14:22 server-1 systemd[1]: Failed to start Application Service.
May 16 10:14:22 server-1 app[432]: Error: Database connection timeout
May 16 11:05:01 server-1 nginx[890]: [Error] 404 Not Found - /favicon.ico
```

5. df -h (Disk Free)

Displays the amount of disk space available on the file systems. The -h flag makes the output "human-readable" (printing sizes in Megabytes and Gigabytes).

```
user@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           796M  1.6M  794M   1% /run
/dev/sda2       49G   12G   35G  26% /
tmpfs           3.9G    0   3.9G   0% /dev/shm
/dev/sda1       512M   61M  452M  12% /boot/efi
tmpfs           796M   64K  796M   1% /run/user/1000
```

6. free -m (Free Memory)

Shows the total amount of free and used physical memory and swap space in the system. The -m flag displays the values in Megabytes.

```
user@ubuntu:~$ free -m
              total        used        free      shared    buff/cache
available
Mem:          7956        2341        1450          120         4165
5120
Swap:         2048           0         2048
```

7. uptime (System Uptime & Load)

Tells you how long the system has been running, how many users are currently logged on, and the system load averages for the past 1, 5, and 15 minutes.

```
user@ubuntu:~$ uptime
12:00:45 up 14 days,  3:22,  2 users,  load average: 0.12, 0.05, 0.01
```

8. ping (Packet Internet Groper)

Tests network connectivity between your machine and another host. The `-c 4` flag limits the test to 4 packets.

```
user@ubuntu:~$ ping -c 4 google.com
PING google.com (142.250.190.46) 56(84) bytes of data.
64 bytes from fra16s51-in-f14.1e100.net: icmp_seq=1 ttl=116 time=14.2 ms
64 bytes from fra16s51-in-f14.1e100.net: icmp_seq=2 ttl=116 time=13.8 ms
64 bytes from fra16s51-in-f14.1e100.net: icmp_seq=3 ttl=116 time=14.5 ms
64 bytes from fra16s51-in-f14.1e100.net: icmp_seq=4 ttl=116 time=13.9 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.821/14.102/14.512/0.264 ms
```

9. chmod (Change Mode)

Changes the access permissions of a file or directory. Here, we add execute (+x) permissions to a python script.

```
# Before changing permissions

user@ubuntu:~/scripts$ ls -l backup.sh
-rw-r--r-- 1 user user 450 May 16 11:30 backup.sh

# Add execute permission

user@ubuntu:~/scripts$ chmod +x backup.sh

# After changing permissions

user@ubuntu:~/scripts$ ls -l backup.sh
-rwxr-xr-x 1 user user 450 May 16 11:30 backup.sh
```

10. history (Command History)

Displays the list of commands you have previously typed in the terminal session. (Output truncated for brevity).

```
user@ubuntu:~$ history | tail -n 5
```

```
1998 df -h  
1999 free -m  
2000 uptime  
2001 ping -c 4 google.com  
2002 history | tail -n 5
```